Bird Species Identification from an Image

Machine Learning Mini-Project Write-up | Team 06 | Project 89
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1. Problem Statement

Bird species recognition is a multi-class classification problem. This project studies whether a bird's annotated visual attributes can be used to predict its species. The work follows the reference study "Bird Species Identification from an Image": first implementing baseline classifiers and comparing their results with the values reported in the study, then adding one extra model as an extension. In the current implementation, the classifiers consume structured attributes rather than raw image pixels; therefore, the notebook does not directly classify a newly uploaded photograph.

2. Dataset Details

The project uses the CUB-200-2011 (Caltech–UCSD Birds) dataset. The dataset contains 11,788 bird images across 200 species and provides 312 binary visual attributes for each image. The attribute annotation file `attributes/image_attribute_labels.txt` records image IDs, attribute IDs and presence flags. The file `image_class_labels.txt` maps image IDs to species class IDs. The annotation file contains 3,677,856 rows.

3. Approach and Methodology

The annotation data is read as whitespace-separated text. The image ID, attribute ID and presence flag are retained, then the annotation rows are pivoted to form one row per image with 312 attribute columns. These features are merged with the species labels. The feature matrix is denoted by X and the target class labels by y. A stratified 70:30 train/test split is created with `random_state=42`, giving 8,251 training samples and 3,537 test samples. The same split is used for the baseline models and the XGBoost extension to support a fair comparison.

4. Implementation and Results

Five baseline classifiers are implemented in the notebook. Their current test accuracies are shown below. These are preliminary results from our implementation and should not be described as a completed reproduction of every experiment in the reference study.

Model	Notebook test accuracy
Bernoulli Naive Bayes	**45.60%**
K-Nearest Neighbors (k = 5)	**33.50%**
Decision Tree	**26.89%**
Random Forest (100 trees)	**46.59%**
Linear Discriminant Analysis (LDA)	**46.76%**

For the extension, an XGBoost classifier is trained on the same 312 attributes and train/test split. The class labels are mapped to integer values from 0 to 199 for training. The configuration is `n_estimators=300`, `max_depth=10`, `learning_rate=0.1`, `subsample=0.8`, `colsample_bytree=0.8`, `random_state=42` and `eval_metric='mlogloss'`. The reported XGBoost test accuracy is **49.39%** (0.4939).

5. Discussion

Among the five current baseline implementations, Linear Discriminant Analysis records the highest test accuracy at 46.76%, closely followed by Random Forest at 46.59%. XGBoost reaches 49.39% on the same holdout set and is the strongest result among the models listed in this write-up. These values describe the current notebook outputs; the reference study reports different values for its baseline and feature-transformation experiments. A numerical difference alone does not establish an error or a successful reproduction. The remaining reproduction work should verify the reference study's exact preprocessing, model settings and PCA/feature-selection procedures before drawing stronger conclusions.

6. Conclusion

This project builds an attribute-based pipeline for classifying 200 bird species using the CUB-200-2011 dataset. The annotations are transformed into a 312-feature matrix, species labels are joined to the features, and a stratified train/test split is used to evaluate multiple classifiers. The current baseline results provide an initial comparison, while XGBoost extends the study and achieves 49.39% test accuracy. The results suggest that classifier choice influences performance on this dataset. Future work should finish reproducing the reference paper's full set of experiments, document any differences in preprocessing and hyperparameters, and report additional metrics such as macro-F1 and a confusion matrix if required. Since the current system uses annotated attributes, classifying a raw photograph would require an additional image-to-attribute component or a separate image-based model.

References

[1] Aditya Bhandari, Ameya Joshi and Rohit Patki, "Bird Species Identification from an Image" (reference study).
[2] CUB-200-2011 dataset, Caltech–UCSD Birds.
[3] Scikit-learn documentation for classification algorithms and evaluation metrics.
[4] XGBoost documentation.

Note: Model accuracies above are taken from the provided notebook/output. The reference paper's full reproduction is still in progress; results should be labelled accordingly in the final submission.